

① 1. let $f_1, \dots, f_m: C \rightarrow \mathbb{R}$ be convex functions
 $\& \lambda_1, \dots, \lambda_m \in \mathbb{R}_+$

$$\text{let } g(u) = \sum_{i=1}^m \lambda_i f_i(u) \quad u, v \in C, \alpha \in [0, 1]$$

$$\begin{aligned} g(\alpha u + (1-\alpha)v) &= \sum_{i=1}^m \lambda_i f_i(\alpha u + (1-\alpha)v) \\ &\leq \sum_{i=1}^m \lambda_i (\alpha f_i(u) + (1-\alpha)f_i(v)) = \alpha \cdot \sum_{i=1}^m \lambda_i f_i(u) + (1-\alpha) \cdot \sum_{i=1}^m \lambda_i f_i(v) \\ &\quad \text{each } f_i \text{ is convex} \\ &= \alpha g(u) + (1-\alpha)g(v) \end{aligned}$$

2. take $f(x) = x^2$ $g(x) = x^2 - 1$ $f'' = g'' = 1 \geq 0$

$$h = f \circ g = (x^2 - 1)^2 = x^4 - 2x^2 + 1$$

$$h'' = 12x^2 - 4 \quad \text{note that } h''(0) = -4$$

$\Rightarrow h$ is not convex

1.2

$$\textcircled{3} \quad x \in \mathbb{R}^d \quad y \in \{\pm 1\}$$

$$f(w, b) = \max_{x, y} (w, b) = \max(0, 1 - y(x^T w + b))$$

$$\text{let } \alpha \in [0, 1] \quad (w, b), (w', b') \in \mathbb{R}^d \times \mathbb{R}$$

$$\text{define } h(w, b) = 0, \quad g(w, b) = 1 - y(x^T w + b)$$

$$\text{note } f(w, b) = \max\{h(w, b), g(w, b)\}$$

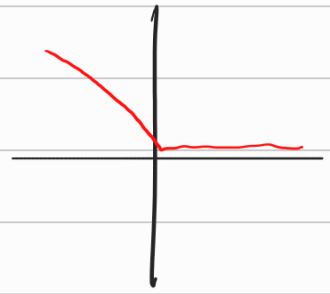
We know from calculus that the max of 2 convex functions is a convex function

h is convex as a constant function

We will show g is convex

Note that $g(w, b) = (-yx)^T w + b + 1$ is affine (linear + const) therefore $\nabla^2 g(w, b) = 0 \succeq 0$
So the Hessian is PSD $\Rightarrow g$ is convex

④ the hinge loss
can be split into 3 regions



• When $y(x^T w + b) > 1$

then $\ell_{x,y}^{\text{hinge}} = 0$ so $\partial \ell_{x,y}^{\text{hinge}} = \{\nabla \ell_{x,y}^{\text{hinge}}\} = \{0\}$

• When $y(x^T w + b) < 1$ so $\ell_{x,y}^{\text{hinge}} = 1 - y(x^T w + b)$

& differentiable so $\partial \ell_{x,y}^{\text{hinge}} = \{\nabla \ell_{x,y}^{\text{hinge}}\}$

$$\nabla \ell_{x,y}^{\text{hinge}} = \begin{pmatrix} \frac{\partial \ell}{\partial w} \\ \frac{\partial \ell}{\partial b} \end{pmatrix} = \begin{pmatrix} -yx^T \\ -y \end{pmatrix}$$

• When $y(x^T w + b) = 1$ we can take any vector under the graph & we can choose one of the gradients we found on the first two cases. Since they are under the graph, for example 0 will always be under the graph of ℓ^{hinge}

⑤ let $f(x) = \sum_{k=1}^m f_k(x)$ by the definition of the subgradient for each convex f_k and any $V \in \mathbb{R}^d$ the following holds:

$$f_k(V) \geq f_k(x) + g_k^T (V - x)$$

$$\Rightarrow \sum_{k=1}^m f_k(V) \geq \sum_{k=1}^m (f_k(x) + g_k^T (V - x))$$

$$\sum f_k(V) \geq \sum f_k(x) + \sum g_k^T (V - x)$$

$$\sum f_k(v) \geq \sum f_k(x) + \left(\sum g_k \right)^T (v-x)$$

$$f(v) \geq f(x) + \left(\sum g_k \right)^T (v-x)$$

this is the definition of the subgradient

$$(6) \quad f(w, b) = \frac{1}{m} \sum_{i=1}^m \lim_{\lambda \rightarrow 0} (v_i, b) + \frac{\lambda}{2} \|w\|^2$$

we can use the linearity property from Q5

$-\frac{\lambda}{2} \|w\|^2$ is convex & smooth so the subgradient is the gradient

$$\nabla_w \frac{\lambda}{2} \|w\|^2 = \lambda w$$

$$\nabla_b \frac{\lambda}{2} \|w\|^2 = 0$$

for each sample i define the indicator

$$I_i = \begin{cases} 1 & \text{if } y_i (v_i^T w + b) < 1 \\ 0 & \text{else} \end{cases}$$

from Q4 the subgradients are $-y_i v_i I_i$ and $-y_i I_i$ for b

$$\Rightarrow \begin{aligned} g_w &= \lambda w - \frac{1}{m} \sum_{i=1}^m (I_i \cdot y_i \cdot x_i) \\ g_b &= -\frac{1}{m} \sum_{i=1}^m (I_i \cdot y_i) \end{aligned} \Rightarrow \begin{pmatrix} g_w \\ g_b \end{pmatrix}$$

$$7 \quad \textcircled{a} \quad \frac{\partial L}{\partial \hat{y}} = y^1 - y \quad \text{since} \quad \hat{y} = w^T h + c$$

$$\frac{\partial L}{\partial h} = (y^1 - y)w, \quad \frac{\partial L}{\partial c} = y^1 - y$$

$$b) \quad \frac{\partial y^1}{\partial w} = \frac{\partial}{\partial h} (w^T h + c) = w$$

$$1. \quad \frac{\partial L}{\partial b} = \frac{\partial L}{\partial h} \cdot \frac{\partial h}{\partial b}$$

$$\frac{\partial h}{\partial b} = \sigma'(wx + b) = \sigma'(wx + b) \cdot 1$$

$$\frac{\partial L}{\partial b} = ((y^1 - y)w) \cdot \sigma'(wx + b)$$

$$2. \quad \frac{\partial L}{\partial w} = \frac{\partial L}{\partial h} \cdot \frac{\partial h}{\partial w}$$

$$\frac{\partial h}{\partial w} = \sigma'(wx + b) \cdot x$$

c) Parts requiring strings

x -input vector used in $w^T x + b$, L (used in $\frac{\partial L}{\partial h}$), y^1

$$8 \quad \textcircled{a} \quad L_i(z) = -\log(z_i)$$

$$\underline{j=i}: \quad \frac{\partial L_i}{\partial z_i} = \frac{\partial}{\partial z_i} (-\log(z_i)) = -\frac{1}{z_i}$$

$$\underline{j \neq i}: \quad \frac{\partial L_i}{\partial z_j} = 0$$

$$\textcircled{b} \quad J_z(L_i) = [0, \dots, -\frac{1}{z_i}, \dots, 0] = -\frac{1}{z_i} e_i^T$$

$$\textcircled{b} \quad J_z(l_i) = \frac{1}{z_i} e_i^T, \quad J_x(s) = \text{diag}(S(x)) - S(x)S(x)^T$$

$$J_x(l_i \circ s) = J_z(l_i) \cdot J_x(s) \Rightarrow \left(\frac{1}{z_i} e_i^T \right) (\text{diag}(z) - zz^T)$$

$$\Rightarrow \underbrace{\left(\frac{1}{z_i} e_i^T \text{diag}(z) \right)}_I - \underbrace{\left(\frac{1}{z_i} e_i^T \right)}_{\overline{A}} zz^T$$

$$\textcircled{I} \quad \frac{1}{z_i} e_i^T \text{diag}(z) = z_i e_i^T \cdot \left(\frac{1}{z_i} \right) = e_i^T$$

$$\textcircled{II} \quad \frac{1}{z_i} e_i^T zz^T = \frac{1}{z_i}$$

$$J_x(l_i \circ s) = z^T - e^T$$